

Industrial Employee Safety and Occupational Health Management System

DSA0101 - OOPS with C++ | Slot D | Capstone Project | Team 8

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Abstract

- Industrial workplaces such as factories, warehouses, and construction sites face constant risks of accidents, injuries, and health hazards caused by unsafe conditions, hazardous gases, machinery, and lack of real-time monitoring.
- Manual safety inspection and paper-based incident reporting are slow, error-prone, and reactive rather than preventive.
- This project proposes an Industrial Employee Safety and Occupational Health Management System that digitizes safety compliance, tracks employee health records, monitors hazard zones, and automates incident reporting and alerts.
- The system is built using Object-Oriented Programming concepts in C++ to model employees, safety equipment, hazard zones, and incident records as reusable, maintainable classes.
- The goal is to reduce workplace accidents, ensure regulatory compliance, and improve response time during emergencies.

Literature Survey (Latest Papers)

Year	Title / Authors	Key Contribution
2024-25	Smart IoT System for Real-Time Worker Safety Monitoring in Dead Zones (IEEE, 2025)	Wearable sensors with ML-based fall detection and location tracking in connectivity-limited zones.
2024-25	Comprehensive IoT-Based Monitoring System for Worker's Health and Safety in High-Risk Environments (IEEE, 2024)	Real-time SpO2, heart rate, gas and temperature monitoring with dashboard alerts for hazardous zones.
2023-24	Shah, I.A. & Mishra, S. — AI in Advancing Occupational Health and Safety (J. Occupational Health, 2024)	Reviews how AI/ML is applied to predict and prevent workplace hazards and improve OHS decision-making.
2023-24	IoT-Enhanced Workplace Safety for Real-Time Monitoring and Hazard Detection (IEEE, 2024)	Sensor network architecture for environmental and wearable monitoring with centralized hazard analytics.
2022-23	IoT Based Wearable Device for Workers in Industrial Scenarios (IEEE, 2023)	ESP-32 based wearable tracking gas exposure, pulse rate, and body temperature with emergency alert button.
2022-23	IoT based Smart Bracelet for Worker Safety and Health Monitoring (IWAIP, IEEE, 2023)	Accelerometer-based vibration monitoring for machine operators with mobile/web notifications.

Issues in the Existing Model

- Safety inspections and incident reports are largely manual and paper-based, causing delays in identifying hazards.
- No centralized system to track employee health history, PPE compliance, and training records together.
- Hazardous zones (gas leaks, high temperature, restricted machinery areas) are not monitored in real time.
- Emergency alerts depend on manual reporting, increasing response time during accidents.
- Existing IoT-based solutions (from literature) focus mainly on sensor hardware and lack an integrated, extensible software backend to manage employees, incidents, and compliance records.
- Lack of role-based access for supervisors, safety officers, and employees in a single management system.

Proposed Model / Solution

- An Industrial Employee Safety and Occupational Health Management System developed using Object-Oriented Programming (C++) principles: classes, inheritance, polymorphism, and encapsulation.
- Employee Module: stores employee profile, health records, and training/certification status.
- Hazard Zone Module: models zones with risk levels and links sensor/alert data to affected employees.
- Incident Reporting Module: digitizes accident/near-miss reporting with timestamps and severity classification.
- PPE & Compliance Module: tracks issuance and compliance status of safety equipment per employee.
- Alert & Notification Module: automatically raises alerts when hazard thresholds are breached or incidents are logged.
- Design emphasizes modularity and reusability so new hazard types or departments can be added without redesigning the system.

System Modules

- **Employee Management Module**
 - Add, update, and view employee details, department, and role
- **Health & Medical Records Module**
 - Maintains periodic health checkup data and flags at-risk employees
- **Hazard Zone Monitoring Module**
 - Classifies zones by risk level and logs sensor-based hazard events
- **Incident & Accident Reporting Module**
 - Records incidents with cause, severity, and corrective action taken
- **PPE & Compliance Tracking Module**
 - Tracks safety equipment issued and compliance verification
- **Alert & Notification Module**
 - Sends real-time alerts to supervisors during hazard or incident events

Conclusion

- The proposed Industrial Employee Safety and Occupational Health Management System replaces manual, fragmented safety processes with a unified, object-oriented software solution.
- By integrating employee records, hazard monitoring, incident reporting, and PPE compliance into one system, organizations can respond faster to emergencies and reduce workplace accidents.
- The modular OOP-based design allows the system to be extended in future with IoT sensor integration and predictive analytics for proactive hazard prevention.
- Overall, the system contributes to a safer, more compliant, and more accountable industrial workplace.

References

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